



Department of Statistics
and Data Science
Faculty of Science

Tutorial 7

Multicollinearity

ST3131 AY25/26 Sem2

2026 March 25

0 What is multicollinearity?

Definition 1 (Multicollinearity). We fit a k regressors MLR $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ where

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}_{n \times 1}, \quad \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix}_{p \times 1} \quad \text{and} \quad \mathbf{X} = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1k} \\ 1 & x_{21} & x_{22} & \cdots & x_{2k} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nk} \end{bmatrix}_{n \times p}.$$

Model assumptions {
(1) independence
(2) normality
(3) linearity
(4) constant variance



✗ Testing results

✓ Computation

0 What is multicollinearity?

Definition 1 (Multicollinearity). We fit a k regressors MLR $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ where

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}_{n \times 1}, \quad \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix}_{p \times 1} \quad \text{and} \quad \mathbf{X} = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1k} \\ 1 & x_{21} & x_{22} & \cdots & x_{2k} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nk} \end{bmatrix}_{n \times p}.$$

If we find linear combinations in the columns of $\mathbf{X}$, for example

$$\mathbf{X}_1 \approx c_2 \mathbf{X}_2 + c_3 \mathbf{X}_3 + \cdots + c_p \mathbf{X}_p \quad \longrightarrow \quad \boxed{\text{Bad computation}}$$

we call the *multicollinearity* is present in this dataset.

0 What is multicollinearity?

Q1 Bad computation

Q2 What caused it?

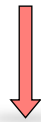
Q3 A deeper look at bad computation

Q1 Bad computation

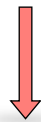
$$\mathbf{X}_1 \approx c_2 \mathbf{X}_2 + c_3 \mathbf{X}_3 + \cdots + c_p \mathbf{X}_p$$

Question 1. What problem does it bring?

$$\mathbf{X}_1 = c_2 \mathbf{X}_2 + c_3 \mathbf{X}_3 + \cdots + c_p \mathbf{X}_p$$



$$\mathbf{X} = \begin{bmatrix} \mathbf{1} & \mathbf{X}_1 & \mathbf{X}_2 & \cdots & \mathbf{X}_p \end{bmatrix} \text{ not of full column rank} \quad \Leftrightarrow \quad \mathbf{X}'\mathbf{X} \text{ has eigenvalue } 0$$
$$\Leftrightarrow \text{rank}(\mathbf{X}'\mathbf{X}) = \text{rank}(\mathbf{X}) < p$$



$$\mathbf{X}'\mathbf{X} \text{ not invertable} \quad \longrightarrow \quad \hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'\mathbf{y} \text{ not computable}$$

$$\text{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2 (\mathbf{X}'\mathbf{X})^{-1} \text{ not computable}$$

Q1 Bad computation

Question 1. What problem does it bring?

$\mathbf{X}_1 = c_2\mathbf{X}_2 + c_3\mathbf{X}_3 + \cdots + c_p\mathbf{X}_p \longrightarrow \mathbf{X}'\mathbf{X} \text{ has eigenvalue } 0 \longrightarrow \hat{\boldsymbol{\beta}} \quad \text{Var}(\hat{\boldsymbol{\beta}})$
not computable

$\mathbf{X}_1 \approx c_2\mathbf{X}_2 + c_3\mathbf{X}_3 + \cdots + c_p\mathbf{X}_p \longrightarrow \mathbf{X}'\mathbf{X} \text{ has eigenvalue } \textit{nearly } 0$
 $\longrightarrow \hat{\boldsymbol{\beta}} \quad \text{Var}(\hat{\boldsymbol{\beta}})$
unstable

Question 2(a). Recall 4 causes for *multicollinearity* in lecture notes.

- **Data collection method** $\begin{cases} \mathbf{X}_1 \leftarrow \text{exercise hr} \\ \mathbf{X}_2 \leftarrow \text{protein intake} \end{cases}$ Interview in the GYM!
- **Constraint in the model** $\begin{cases} \mathbf{X}_1 \leftarrow \text{assignment score} \\ \mathbf{X}_2 \leftarrow \text{midterm score} \\ \mathbf{X}_3 \leftarrow \text{final score} \end{cases}$ $\mathbf{X}_1 + \mathbf{X}_2 + \mathbf{X}_3 = 100$
- **Model specification** $\mathbf{X}_1 \quad \mathbf{X}_2 \leftarrow (\mathbf{X}_1)^2 \quad \mathbf{X}_3 \leftarrow (\mathbf{X}_1)^3$
- **Over-defined model** sample size $n < p$ regressors num.

Question 2(c). Propose remedy to fix them.

- **Data collection method**

x_1 = yr of work and x_2 = salary,

and we only sample from senior engineers

- **Constraint in the model**

$x_1 = \text{N}_2 \%$ $x_2 = \text{O}_2 \%$ $x_3 = \text{other } \%$

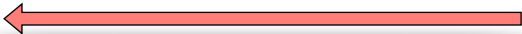
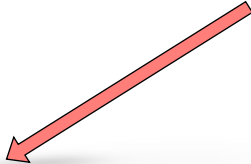
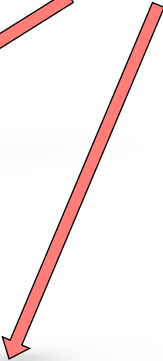
- **Model specification**

To fix the non-linearity, we chose $y \sim x + x^2$

- **Over-defined model**

For a $p = 1,000,000$ DNA gene study we have $n = 8$ patient.

Question 2(c). Propose remedy to fix them.

- **Data collection method**  Sample design
 $x_1 = \text{yr of work}$ and $x_2 = \text{salary}$,
and we only sample from senior engineers
- **Constraint in the model**  Just remove one regressor
 $x_1 = \text{N}_2 \%$ $x_2 = \text{O}_2 \%$ $x_3 = \text{other } \%$ Ridge / PCA (later in this mod)
- **Model specification**
To fix the non-linearity, we chose $y \sim x + x^2$ 
- **Over-defined model**
For a $p = 1,000,000$ DNA gene study we have $n = 8$ patient. 

0 What is multicollinearity?

Q1 Bad computation

Q2 What caused it?

Q3 A deeper look at bad computation

Q3 A deeper look at bad computation

Question 3(a). Explain why high $\text{sampCor}(\mathbf{X}_1, \mathbf{X}_2)$ leads to large $\text{Var}(\hat{\beta})$

$\mathbf{X}_1 \approx c_2 \mathbf{X}_2 + c_3 \mathbf{X}_3 + \dots + c_p \mathbf{X}_p \longrightarrow \mathbf{X}'\mathbf{X}$ has eigenvalue *nearly* 0 $\longrightarrow \hat{\beta}$ and $\text{Var}(\hat{\beta})$ unstable

```
# Let's use viscosity.csv as example
y = read.csv("viscosity.csv")$visc
x1 = read.csv("viscosity.csv")$temp
x2 = 0.99 * x1 + 40 + rnorm(n = 8, sd = 0.01)
# try print cor(x1, x2)
```

```
X = cbind(1, x1, x2) # Create design matrix
eigen(t(X) %*% X)$values # Eigen-decompose
solve(t(X) %*% X) # Invert matrix
```

```
> eigen(t(X) %*% X)$values
[1] 1.16e+05    4.68e+02    2.65e-07
> solve(t(X) %*% X)
      [,1]      [,2]      [,3]
[1,] 3.77e+06  9.34e+04 -9.34e+04
[2,] 9.34e+04  2.31e+03 -2.33e+03
[3,] -9.34e+04 -2.33e+03  2.35e+03
```

Q3 A deeper look at bad computation

Question 3(b). Suppose $\mathbf{X}_1, \dots, \mathbf{X}_p$ are standardized, i.e. $\text{sampVar}(\mathbf{X}_i) = 1$ for all i . If $\mathbf{X}_p \sim \mathbf{X}_1 + \dots + \mathbf{X}_{p-1}$ has R_p^2 close to 1, what happens if we fit $\mathbf{y} \sim \mathbf{X}_1 + \dots + \mathbf{X}_p$?

$$\text{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2 (\mathbf{X}'\mathbf{X})^{-1}$$

$$\begin{bmatrix} \text{Var}(\hat{\beta}_1) & \text{Cov}(\hat{\beta}_1, \hat{\beta}_2) & \cdots & \text{Cov}(\hat{\beta}_1, \hat{\beta}_p) \\ \text{Cov}(\hat{\beta}_2, \hat{\beta}_1) & \text{Var}(\hat{\beta}_2) & \cdots & \text{Cov}(\hat{\beta}_2, \hat{\beta}_p) \\ \vdots & \vdots & \ddots & \vdots \\ \text{Cov}(\hat{\beta}_p, \hat{\beta}_1) & \text{Cov}(\hat{\beta}_p, \hat{\beta}_2) & \cdots & \text{Var}(\hat{\beta}_p) \end{bmatrix}$$
$$\sigma^2 \begin{bmatrix} C_{11} & C_{12} & \cdots & C_{1p} \\ C_{21} & C_{22} & \cdots & C_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ C_{p1} & C_{p2} & \cdots & C_{pp} \end{bmatrix}$$

we use C to denote $(\mathbf{X}'\mathbf{X})^{-1}$
and C_{ij} for its elements.

Q3 A deeper look at bad computation

Question 3(b). Suppose $\mathbf{X}_1, \dots, \mathbf{X}_p$ are standardized, i.e. $\text{sampVar}(\mathbf{X}_i) = 1$ for all i . If $\mathbf{X}_p \sim \mathbf{X}_1 + \dots + \mathbf{X}_{p-1}$ has R_p^2 close to 1, what happens if we fit $\mathbf{y} \sim \mathbf{X}_1 + \dots + \mathbf{X}_p$?

$$\text{Var}(\hat{\beta}_p) = \sigma^2 C_{pp} = \frac{\sigma^2}{1 - R_p^2} \xrightarrow{\text{as } R_p^2 \rightarrow 1} \infty$$

↑

Hint: $C_{pp} = \frac{1}{1 - R_p^2}$

0 What is multicollinearity?

$$\mathbf{X}_1 \approx c_2 \mathbf{X}_2 + c_3 \mathbf{X}_3 + \cdots + c_p \mathbf{X}_p$$

Q1 Bad computation

$\hat{\boldsymbol{\beta}}$ and $\text{Var}(\hat{\boldsymbol{\beta}})$ unstable

Q2 What caused it?

4 causes in the lecture

Q3 A deeper look at bad computation